

NUMERICAL ANALYSIS

Polynomial:

An expression of the form $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n$ where n is a positive integers and $a_0, a_1, a_2, \dots, a_n$ are real constants, is called a polynomial in x of n th degree if $a_0 \neq 0$.

Algebraic Equation:

An equation of the form $f(x) = 0$, where $f(x)$ is a polynomial of n th degree, is called algebraic equation or polynomial equation in x of n th degree. For example $x^2 + x + 1 = 0$.

Transcendental Equation:

An equation which is not algebraic equation is called transcendental equation.

For example $\sin x + e^x = 0$.

Some Fundamental Theorems:

- (i) Every equation of the form $f(x) = 0$ has at least one root, either real or complex.
- (ii) Every polynomial equation of n th degree has n and only n roots.
- (iii) If $f(x) = 0$ is a polynomial equation and if $f(a), f(b)$ are of different signs, then $f(x) = 0$ must have at least one real root in between a and b .
- (iv) If $f(x) = 0$ is an equation of odd degree, then it has at least one real root whose sign is opposite to that of the last term.
- (v) If $f(x) = 0$ is an equation of even degree whose last term is $-ve$, then it has at least one $+ve$ root and at least one $-ve$ root.
- (vi) If $f(x) = 0$ has no real root between a and b , ($a < b$), then $f(a)$ and $f(b)$ are of the same sign.
- (vii) In a polynomial equation $f(x) = 0$ with real coefficient, imaginary roots occur in conjugate pairs.
- (viii) In a polynomial equation $f(x) = 0$ with rational coefficients, irrational roots in the form of quadratic surds occur in conjugate pairs.

Relationships Between Roots And Coefficients:

The following table shows the information of relationships between roots and coefficients of different orders equations.

Equation	General Form	Roots	Relationships
Quadratic	$ax^2 + bx + c = 0$, where $a \neq 0$	α, β	$\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$
Cubic	$ax^3 + bx^2 + cx + d = 0$ Where $a \neq 0$	α, β, γ	$\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$, $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
Quartic	$ax^4 + bx^3 + cx^2 + dx + e = 0$, Where $a \neq 0$	$\alpha, \beta, \gamma, \delta$	$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$, $\alpha\beta\gamma\delta = \frac{e}{a}$, $a\beta + \alpha\gamma + \alpha\delta + \beta\alpha$ $+ \beta\delta + \gamma\delta = \frac{c}{a}$, $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta +$ $\beta\gamma\delta = -\frac{d}{a}$

Nature of Roots of Quadratic Equation $ax^2 + bx + c = 0$, $a \neq 0$:

The following table shows the information of nature of roots of a quadratic equation.

Condition	Nature
$b^2 - 4ac = 0$	Roots are equal and rational
$b^2 - 4ac > 0$	Roots are different and rational
$b^2 - 4ac > 0$ and is not a perfect square	Roots are different and Irrational
$b^2 - 4ac < 0$	Roots are conjugate complex numbers
$b^2 = \frac{9ac}{2}$	One root is double the other
$b^3 + a^2c + ac^2 = 3abc$	One root is square of the other
$b = 0$	One root is additive inverse of the other
$a - c = 0$	One roots is multiplication inverse of the other

Nature of Roots of Cubic Equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$:

The following table shows the information of nature of roots of a 3rd degree equation.

Condition	Nature
$2b^3 - 9abc + 27da^2 = 0$	Roots are in arithmetical progression
$ac^3 - b^3d = 0$	Roots are in geometrical progression
$2c^3 - 9bcd + 27d^2a = 0$	Roots are in harmonic progression

Numerical Methods for Solving Equations:

The following table shows the names of numerical methods with their rate of convergence for solving algebraic and transcendental equations.

Name	Rate of Convergence
Bisection method	1.00
Regula Falsi Method	1.00
Newton Raphson Method	2.00
Secant Method	1.67

Methods for Solving System of Linear Algebraic Equations:

The following table shows the names of methods for solving system of equations.

Direct Methods	Iterative Methods
Matrix Inversion Method Cramer's Rule Gauss Elimination Method Gauss-Jordan Method LU - Decomposition Method Crout's Method	Gauss-Jacobi Method Gauss-Seidel Method Relaxation Method

Difference Operators:

The following table shows the information of difference operators.

Forward Difference operator	$\Delta f(x) = f(x+h) - f(x)$
Backward Difference operator	$\nabla f(x) = f(x) - f(x-h)$
Central difference operator	$\delta f(x) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right)$
Shift operator	$Ef(x) = f(x+h)$

Averaging operator	$\mu f(x) = \frac{1}{2} \left[f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right) \right]$
Differential operator	$Df(x) = \frac{d}{dx} f(x)$
Unit operator	1. $f(x) = f(x)$

Relation Between Operators:

Following are the relations between difference operators.

- (i) $\Delta = E - 1$
- (ii) $E = (1 - \nabla)^{-1}$
- (iii) $\delta = E^{1/2} \nabla = E^{-1/2} \Delta$
- (iv) $U = \frac{1}{2} (E^{1/2} + E^{-1/2})$
- (v) $E = e^{hD}$
- (vi) $D = \frac{1}{h} \sin^{-1} (\mu\delta)$

Interpolation:

The process of finding the value of function inside the given range of arguments is called interpolation.

Extrapolation:

The process of finding the value of a function outside the given range of arguments is called extrapolation.

Interpolation Formulae:

The following table shows the information of different interpolation formulae that are used in interpolation..

Geogory – Newton Forward Interpolation Formula
$y_p = y_0 + p \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \dots + \frac{P(P-1)(P-2) \dots [P-(n-1)]}{n!} \Delta^n y_0$
Gregory – Newton Backward Interpolation Formula
$y_p = y_n + p \nabla y_n + \frac{P(P-1)}{2!} \nabla^2 y_n + \dots + \frac{P(P-1)(P-2) \dots (p+n-1)]}{n!} \nabla^n y_n$
Gauss Forward Interpolation Formula
$y_p = y_0 + p \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_{-1} + \dots + \frac{(p+1)(p-1)}{3!} \Delta^3 y_{-1}$ $+ \frac{(P+1)P(P-1)(P-2)]}{4!} \Delta^4 y_{-2} + \dots$

Gauss' Backward Interpolation Formula

$$y_p = y_0 + p\Delta y_{-1} + \frac{(P+1)}{2!} P\Delta^2 y_{-1} + \frac{(P+1)P(P-1)}{3!} \Delta^3 y_{-2} \\ + \frac{(P+2)(P+1)P(P-1)}{4!} \Delta^4 y_{-2} + \dots$$

Stirling's Formula

$$y_p = y_0 + pu \delta y_0 + \frac{P^2}{2!} \delta^2 y_0 + \frac{P(P^2-1)}{3!} u\delta^3 y_0 + \frac{P^2(P^2-1)}{4!} \delta^4 y_0 + \dots$$

Bessel's Formula

$$y_p = \frac{1}{2}(y_0 + y_1) + \left(P - \frac{1}{2}\right) \Delta y_0 + \frac{(P-1)}{2!} \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right] + \dots$$

Laplace-Everett's Formula

$$y_p = ((1-p)y_0 + py_1) - \frac{P(P-1)(P-2)}{3!} \Delta^2 y_{-1} + \frac{(P+1)P(P-1)}{3!} \Delta^2 y_0 + \dots$$

Numerical Differentiation:

The process of computing the derivative or derivatives of the function $f(x)$ at some values of x from the given set of values is called Numerical differentiation.

The following table shows the information of some formulae that are used in numerical differentiation.

Newton's Forward difference Formula	$\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{2P-1}{2!} \Delta^2 y_0 + \frac{3p^2-6p+2}{3!} \Delta^3 y_0 + \dots \right]$
Newton's Backward difference Formula	$\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_0 + \frac{2P+1}{2!} \nabla^2 y_0 + \frac{3p^2+6p+2}{3!} \nabla^3 y_0 + \dots \right]$
Stirling's Formula	$\frac{dy}{dx} = \frac{1}{h} \left\{ \frac{\Delta y_0 + \Delta y_{-1}}{2} \right\} + p\Delta^2 y_{-1} \\ + \frac{3(P^2-1)}{7} \left\{ \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} + \dots \right\}$

Numerical Integration:

The process of computing $\int_a^b y \, dx$, where $y = f(x)$ is given by a set of tabulated values $[x_i, y_i]$ $i = 0, 1, 2, \dots, n$, $a = x_0$ and $b = x_n$ is called numerical integration.

Newton-Cote's Quadrature Formula:

$$\int_{x_0}^{x_0+nh} f(x) dx = h \left[ny_0 + \frac{n^2}{2} \Delta y_0 + \frac{1}{2} \left\{ \frac{n^3}{3} - \frac{n^2}{2} \right\} \Delta^2 y_0 + \frac{1}{6} \left\{ \frac{n^4}{4} - n^3 + n^2 \right\} \Delta^3 y_0 + \dots \right]$$

Integration Formulae by Newton-Cote's Quadrature Formula:

The following table shows the information of different numerical integration formulae.

$n = 1$	Trapezoidal Rule	$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$
$n = 2$	Simpson's $\frac{1}{3}$ Rule	$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{3} [(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$
$n = 3$	Simpson's $\frac{3}{8}$ Rule	$\int_{x_0}^{x_0+nh} f(x) dx = \frac{3h}{8} [(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})]$
$n = 6$	Weddle's Rule	$\int_{x_0}^{x_0+nh} f(x) dx = \frac{3h}{10} [(y_0 + y_n) + (5y_1 + y_2 + 6y_3 + y_4 + 5y_5) + \dots + 2(2y_{n-6} + 5y_{n-5} + y_{n-4} + 6y_{n-3} + y_{n-2} + 5y_{n-1})]$

Errors in Numerical Methods:

Following table shows the information of errors of different numerical integration formulae.

Trapezoidal Rule	$\frac{-h^3}{12} y''$
Simpson's $\frac{1}{3}$ Rule	$\frac{-h^5}{90} y^{IV}$
Simpson's $\frac{3}{8}$ Rule	$\frac{-3h^5}{80} y^{IV}$
Weddle's Rule	$\frac{-h^7}{140} y^{VI}$

Multiple Choice Questions:

- If α, β are roots of equation, $ax^2 + bx + c = 0$, $a \neq 0$, then which of the following is not correct.
 - $\alpha + \beta = \frac{-b}{a}$
 - $\alpha\beta = \frac{c}{a}$
 - $\alpha\beta + \alpha = ac$
 - none
- If α, β, γ are roots of equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, then which of the following is correct.
 - $\alpha + \beta + \gamma = -\frac{b}{a}$
 - $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$
 - $\alpha\beta\gamma = -\frac{d}{a}$
 - all of these
- If $\alpha, \beta, \gamma, \delta$ are roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, $a \neq 0$ then which of the following is correct.
 - $\alpha + \beta + \gamma + \delta = \frac{-b}{a}$
 - $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$
 - $\alpha\beta\gamma\delta = \frac{e}{a}$
 - all of these
- If $\alpha, \beta, \gamma, \delta$ are roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$, $a \neq 0$, then which of the following is correct.
 - $\alpha\beta\gamma + \beta\gamma\delta + \alpha\beta\delta = \frac{-d}{a}$
 - $\alpha\beta\gamma + \alpha\gamma\delta + \beta\gamma\delta + \alpha\beta\delta = \frac{-d}{a}$
 - $\alpha\beta\gamma + \beta\gamma\delta + \alpha\gamma\delta = \frac{-d}{a}$
 - none
- If $\alpha + i\beta$ is one root of the equation $f(x) = 0$, then other root of $f(x) = 0$ is
 - α
 - β
 - $\alpha - i\beta$
 - $\beta - \alpha i$
- If $\alpha + \sqrt{\beta}$ is a root of an equation $f(x) = 0$, where α, β are rational numbers and $\beta \neq 0$, then other root of the equation $f(x) = 0$ is
 - α
 - $\alpha + i\beta$
 - $\alpha - \sqrt{\beta}$
 - $\sqrt{\alpha\beta}$
- The root of the equation $x^3 + 6x + 20 = 0$ is $1 - 3i$. Then which of the following is also a root.
 - 2
 - $1 + 3i$
 - $1 + \sqrt{3}$
 - $3i$
- The root of the equation $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ is $3 + 2\sqrt{2}$. Then which of the following is also a root.
 - $1 - 2\sqrt{2}$
 - $2 + 3\sqrt{2}$
 - $3 - 3\sqrt{2}$
 - $3 - 2\sqrt{2}$
- One root of the equation $3x^5 - 4x^4 - 42x^3 + 56x^2 + 27x - 36 = 0$ is $\sqrt{2} + \sqrt{5}$, then other root is
 - $\sqrt{2} - \sqrt{5}$
 - $-\sqrt{2} - \sqrt{5}$

- (c) $5\sqrt{2}$ (d) $\sqrt{2} - i\sqrt{5}$
10. Let α be a multiple root of equation $f(x) = 0$ with multiplicity k . Then α is also a multiple root of $f'(x)$ with multiplicity.
- (a) k (b) $k - 1$
(c) $k + 1$ (d) $2k$
11. If $f(x) = 0$ has a multiple (repeated) root α of multiplicity 3, then $f'(x) = 0$ has root α of multiplicity.
- (a) 1 (b) 2
(c) 3 (d) 4
12. The equation $x^2 + 3x + 1 = 0$ is:
- (a) algebraic equation (b) transcendental equation
(c) cubic equation (d) none
13. The equation $x^3 - \log_{10}x + \sin x = 0$ is
- (a) algebraic equation (b) polynomial equation
(c) transcendental equation (d) none
14. The root of the equation $f(x) = 0$ is in interval $[a, b]$ if:
- (a) $f(a) < 0$ (b) $f(a)f(b) < 0$
(c) $f(a) < 0$ and $f(b) < 0$ (d) $f(a) > 0$ and $f(b) > 0$
15. The bisection method is also known as
- (a) Bolzano method (b) Interval halving method
(c) successive method (d) (a) & (b) only
16. In bisection method for $f(x) = 0$ if $f(a) < 0$, $f(b) > 0$ and $f\left(\frac{a+b}{2}\right) < 0$ then root lies between
- (a) a and $\frac{a+b}{2}$ (b) b and $\frac{a+b}{2}$
(c) $2a$ and $a+b$ (d) $2b$ and $a+b$
17. The method of successive approximation is also known as
- (a) secant method (b) iteration method
(c) convergent method (d) none
18. If α be a root of $f(x) = 0$ which is equivalent to $x = \phi(x)$ and I be an interval such that $x = \alpha \in I$, then sequence of iteration $x_0, x_1, x_2, \dots, x_n$ converges to α if $\forall x \in I$.
- (a) $|\phi(x)| < 0$ (b) $|\phi'(x)| < 1$
(c) $|\phi'(x) - \alpha| < 1$ (d) $|\phi'(x)| > 0$
19. The order of convergence of iteration method is:
- (a) 0 (b) 1
(c) 2 (d) 4
20. The method of false position is also unknown as
- (a) bisection method (b) secant method
(c) regula falsi method (d) none
21. The regula falsi method is somewhat similar to
- (a) secant method (b) bisection method

- (c) iteration method (d) none
22. In regula falsi method for the root of $f(x) = 0$ if $f(a) < 0$ and $f(b) > 0$ the next approximation to the root is
- (a) $\frac{f(b) - f(a)}{a + b}$ (b) $\frac{af(b) - bf(a)}{b - a}$
- (c) $\frac{af(b) - bf(a)}{f(b) - f(a)}$ (d) none
23. The Newton-Raphson formula can be derived from
- (a) Binomial theorem (b) Taylor's theorem
- (c) Role's theorem (d) Bisection formula
24. Which of the following is Newton-Raphson formula for the root of equation $f(x) = 0$.
- (a) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, $n = 0, 1, 2, \dots$ (b) $x_{n+1} + \frac{f(x_n)}{f'(x_n)} = x_n$, $n = 0, 1, 2, \dots$
- (c) $x_n - x_{n+1} = \frac{-f(x_n)}{f'(x_n)}$, $n = 0, 1, 2, \dots$ (d) all of these
25. The Newton-Raphson method fails if in the neighbourhood of the root.
- (a) $f(x) = 0$ (b) $f'(x) = 0$
- (c) $f'(x) = 0$ (d) $f'(x) = \infty$
26. The rate of convergence of Newton-Raphson method is:
- (a) 1 (b) 2
- (c) 3 (d) 4
27. Which of the following is iterative method.
- (a) Gauss elimination method (b) Crouts method
- (c) Gauss-seidel method (d) Gauss-Jordan method
28. The relaxation method is
- (a) direct method (b) iterative method
- (c) simple method (d) algebraic method
29. Which of the following is modification of Gauss elimination method.
- (a) Gauss-Seidel method (b) Gauss-Jorden method
- (c) Gauss-Jacobi method (d) Crouts method
30. Method of Factorization is also known as
- (a) triangularization method (b) LU-Decomposition method
- (c) Choleski method (d) all of these
31. Which of the following method is based on the fact that every square matrix can be expressed as a product of a lower triangular matrix and a unit upper triangular matrix.
- (a) Crout's method (b) Choleski method
- (c) LU decomposition method (d) all of these
32. For the solution to a system of linear equations, the system should be diagonally dominant, is condition for
- (a) direct method (b) indirect method
- (c) iterative method (d) (b) and (c) only

33. Which of the following system is diagonally dominant.
- (a) $x_1 + 2x_2 + x_3 = 0$, $x_1 - x_3 = 9$, $x_1 + x_2 + x_3 = 0$
(b) $2x_1 + x_2 + x_3 = 1$, $2x_2 + x_3 = 0$, $x_1 + x_2 = 2$
(c) $x_1 + 3x_2 = 1$, $x_1 + 2x_2 + 4x_3$, $5x_1 + x_2 + x_3 = 0$
(d) none
34. Which of the following is modification of Gauss-Jacobi method.
- (a) Gauss elimination method (b) Gauss Jordan method
(c) Gauss Seidel method (d) none
35. The rate of convergence of Gauss-Seidel is twice that of
- (a) Gauss elimination method (b) Gauss Jordan method
(c) Gauss Jacobi method (d) none
36. Which of the following is forward difference.
- (a) $\Delta f(x) = f(x+h) - f(x)$ (b) $\Delta^2 f(x) = f(x+2h) - 2f(x+h) + f(x)$
(c) $\Delta^3 f(x) = f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)$
(d) all of these
37. Which of the following is true for operator Δ .
- (a) $\Delta(f(x) + g(x)) = \Delta f(x) + \Delta g(x)$ (b) $\Delta^m \Delta^n f(x) = \Delta^{m+n} f(x)$
(c) $\Delta(f(x).g(x)) = f(x) . \Delta g(x)$ (d) $\Delta(kf(x)) = k\Delta f(x)$, k is a constant
38. The value of the independent variable is called:
- (a) domain (b) argument
(c) output (d) amplitude
39. _____ of difference equation is an expression for $y(n)$, which satisfies the given difference equation.
- (a) function (b) solution
(c) equation (d) none
40. A difference equation is a relation between the differences of an unknown function at values of
- (a) argument (b) variable
(c) constant (d) none
41. If in a difference equation y^{n+1} , y^{n+2} etc. occur to the first degree only and are not multiple together, then that difference equation is
- (a) linear (b) quadratic
(c) cubic (d) none
42. The symbol used for central difference is
- (a) Δ (b) δ
(c) μ (d) β

43. The symbol for shift operator is
(a) Δ (b) δ
(c) μ (d) E
44. Which of the following is true.
(a) $\delta = E - 1$ (b) $\delta = 2E + 1$
(c) $\delta = E^{1/2} - E^{-1/2}$ (d) $\delta = E^{1/2} - 2E$
45. Which of the following is a representation for backward difference.
(a) $\nabla f(x) = f(x) - f(x+h)$ (b) $\nabla f(x) = f(x) - f(x-h)$
(c) $\nabla f(x+h) = f(x-h) - f(x)$ (d) $\nabla f(x-h) = f(x+h) - f(x)$
46. The n th difference of a function tabulated at equally spaced intervals is
(a) zero (b) linear
(c) constant (d) one
47. Which of the following is a symbol for shift operator.
(a) Δ (b) μ
(c) E (d) D
48. Which of the following is correct.
(a) $E = (1 - \nabla)^{-1}$ (b) $E = 1 + \Delta$
(c) $\delta = E^{1/2} \nabla$ (d) all of these
49. D is a symbol of
(a) central difference operator (b) shift operator
(c) differential operator (d) difference operator
50. Which of the following is not correct.
(a) $\mu\delta = \frac{\Delta + \nabla}{2}$ (b) $\Delta^2 = (1 + \Delta) \delta^2$
(c) $(1 + \Delta)(1 - \nabla) = 1$ (d) $\mu = \frac{\Delta}{1 + \Delta}$
51. If the interpolation is required near the beginning of the table then which formula is usual full.
(a) Gregory-Newton's forward formula
(b) Stirling's formula
(c) Bessel's formula (d) Laplace formula
52. If interpolation is required near the end of the table then which formula is usualful.
(a) Gregory-Newton's Backward formula
(b) Stirling's formula
(c) Bessel's formula (d) Laplace's formula
53. If interpolation is required near the centre of the table, then which formula is useful.
(a) Bessel's formula (b) Stirling's formula
(c) Laplace–Everett's formula (d) all of these
54. Interpolation with unequal intervals requires.
(a) difference formulae (b) divided difference formulae
(c) central difference formulae (d) none

55. Numerical integration formulae are derived from
(a) Newton-Gragory's formula (b) Newton-Cote's formula
(c) Newton-Raphson formula (d) none
56. The trapezoidal rule derived from New-Cotes quadrature formula by replacing $n =$
(a) 1 (b) 2
(c) 3 (d) 4
57. The Weddle's rule is obtained from quadrature formula by replacing $n =$
(a) 1 (b) 2
(c) 4 (d) 6
58. The error in Trapzoidel rule is of the order.
(a) h^2 (b) h^3
(c) h^4 (d) h^5
59. The error in Trapzoidal rule is
(a) $\frac{-(b-a)h^2}{12} y''(\xi)$ (b) $\frac{-(b-a)h^4}{180} y^{IV}(\xi)$
(c) $\frac{-(b-a)3h^4}{80} y^{IV}(\xi)$ (d) $\frac{-(b-a)h^7}{140} y^{IV}(\xi)$
60. The error in Simpson's $\frac{3}{8}$ rule is of the order.
(a) h^2 (b) h^3
(c) h^4 (d) h^5
61. The error in Weddle's rule is
(a) h^2 (b) h^4
(c) h^6 (d) h^7
62. The method of finding a specific relation $y = f(x)$ for the data to satisfy as accurately as possible is called the
(a) fitting curve (b) curve of equation
(c) regression (d) curve of best fit
63. When curve fitting is done for the straight line $y = a + bx$ by least square method, it has _____ normal equation.
(a) 1 (b) 2
(c) 3 (d) 4
64. Process of estimating the value of dependent variable at an intermediate value is called.
(a) interpolation (b) extrapolation
(c) estimation (d) none
65. The polynomial $y = f(x)$ satisfying the data is called
(a) polynomial (b) extrapolating polynomial
(c) interpolating polynomial (d) manomial
66. Error in Simpson's $\frac{3}{8}$ rule is large as compared to
(a) Trapzoidel rule (b) Simpson's $\frac{1}{3}$ rule
(c) Weddle's rule (d) none

67. Simpson's $\frac{1}{3}$ rule is obtained from general quadrature formula by taking $n =$
(a) 1 (b) 2
(c) 3 (d) 4
68. To apply Simpson's $\frac{3}{8}$ rule, the number of intervals can be
(a) 10 (b) 11
(c) 12 (d) 13
69. Simpson's $\frac{1}{3}$ rule can apply only if given interval is divided into an _____ number of equal interval.
(a) odd (b) even
(c) uncountable (d) none
70. In which of the following method we approximate the curve of solution by the tangent in each interval.
(a) Picard's method (b) Euler's method
(c) R – K method (d) none
71. Which of the following is faster than other in convergence.
(a) Simple Euler's method (b) Euler's method
(c) modified Euler's method (d) Picard's method
72. Which of the following is always convergent.
(a) Bisection method (b) Regula false method
(c) Newton's Raphson (d) (a) & (b) only
73. The Trapezoidal rule approximates the value of a
(a) proper integral (b) improper integral
(c) indefinite integral (d) definite integral
74. The basic idea of relaxation method is to reduce the largest residual to
(a) 1 (b) 2
(c) 0 (d) 5
75. While using relaxation method, which of the following is largest residual for 1st iteration on the system $2x + 3y = 1$, $3x + 2y = -4$.
(a) 4 (b) 3
(c) 2 (d) 1
76. The eigen values of a symmetric matrix are all
(a) rational (b) same
(c) real (d) complex
77. Power method is applicable if the eigen vectors corresponding to eigen values are
(a) same (b) different
(c) linearly independent (d) linearly dependent
78. Power method is applicable if the eigen values are
(a) real (b) rational
(c) real and distinct (d) complex and distinct
79. In a square matrix A, $|\det(A)| =$ product of absolute, values of
(a) diagonal element (b) eigen values of matrix A
(c) trace of A (d) none

-
80. For distinct eigen values, the eigen vectors are orthogonal in a
(a) square matrix (b) scalar matrix
(c) symmetric matrix (d) diagonal matrix
81. All eigen values of a symmetric matrix are
(a) distinct (b) real
(c) complex (d) same
82. Which of the following is predictor-corrector method
(a) Euler's method (b) Milne's method
(c) R – K method (d) Picard's method
83. The number of starting points required for Milne's method are
(a) 2 (b) 3
(c) 4 (d) 5

KEY to MCQ of Numerical Analysis

1	(c)	2	(d)	3	(d)	4	(b)	5	(c)	6	(c)
7	(b)	8	(d)	9	(b)	10	(b)	11	(b)	12	(a)
13	(c)	14	(b)	15	(d)	16	(b)	17	(b)	18	(b)
19	(b)	20	(c)	21	(b)	22	(c)	23	(b)	24	(d)
25	(b)	26	(b)	27	(c)	28	(b)	29	(b)	30	(d)
31	(d)	32	(d)	33	(c)	34	(c)	35	(c)	36	(d)
37	(c)	38	(b)	39	(b)	40	(a)	41	(a)	42	(b)
43	(d)	44	(c)	45	(b)	46	(b)	47	(c)	48	(d)
49	(c)	50	(d)	51	(a)	52	(a)	53	(d)	54	(b)
55	(b)	56	(a)	57	(d)	58	(a)	59	(a)	60	(d)
61	(d)	62	(d)	63	(b)	64	(a)	65	(c)	66	(b)
67	(c)	68	(c)	69	(b)	70	(b)	71	(c)	72	(d)
73	(d)	74	(c)	75	(b)	76	(b)	77	(c)	78	(c)
79	(b)	80	(c)	81	(b)	82	(d)	83	(c)		